

RAAZ DWIVEDI

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ACADEMIC APPOINTMENTS

Chief Technology Officer and Co-founder, Traversal 2024—
Assistant Professor, Operations Research & Information Engineering (ORIE) 2024—
 Field Member: Applied Math, Computer Science, ORIE, Statistics
 Cornell Tech, Cornell University
Visiting Assistant Professor, ORIE, Cornell University Fall 2023
FODSI Postdoctoral Fellow, CS, Statistics, EECS 2021–2023
 Harvard University & Massachusetts Institute of Technology (MIT)
 Advisors: *Prof. Susan Murphy & Prof. Devavrat Shah*

EDUCATION

Ph.D., EECS, University of California (UC), Berkeley 2015–2021
 Advisors: *Prof. Martin Wainwright & Prof. Bin Yu*
 Thesis title: *Principled statistical approaches for sampling and inference in high dimensions*
B. Tech., EE, Indian Institute of Technology (IIT), Bombay, India 2010–2014
 Minors in mathematics, Institute Rank 1

RESEARCH INTERESTS

My research involves a multi-disciplinary approach to data science and brings together ideas from computer science, electrical engineering, and statistics in collaboration with domain experts. I develop statistical machine learning approaches for data-driven personalized decision-making with research across *causal inference, reinforcement learning, Bayesian inference, random sampling, and high-dimensional statistics*.

SELECTED ACHIEVEMENTS & AWARDS

- Delta Rising Professor, Delta 2026
- Blackwell Rosenbluth Award for contributions to Bayesian statistics 2024
- Best Student Paper Award, Statistical Computing & Graphics, American Statistical Association 2022
- Best Presentation Award, Laboratory of Information & Decision Systems (LIDS) Conference, MIT 2022
- Certificate of Distinction and Excellence in Teaching (Q Award), Harvard University 2022
- Foundations of Data Science (FODSI) Postdoctoral Fellowship 2021
- Outstanding Graduate Student Instructor Award, UC Berkeley 2020
- Berkeley Fellowship, the most prestigious fellowship for incoming Ph. D. students 2015
- President of India Gold Medal, IIT Bombay, for the highest GPA in the graduating class 2014
- All India Rank 10 amongst half a million, IIT Joint Entrance Exam 2010

PRE-PRINTS

- (^{*} denotes equal contribution and [†] denotes alphabetical ordering; title is hyperlinked to the online pdf of the paper)
- P1. Albert Gong, Kyuseong Choi, Abhineet Agarwal, Jason Schechner, Ryan Huang, Raj Agrawal, Anish Agarwal, **Raaz Dwivedi**, “ORCA-bench: How ready are language model agents for oncall?”, *arxiv*.2026
- P2. Albert Gong, Annabelle Michael Carrell, **Raaz Dwivedi**, Lester Mackey, “Express language modeling”, *arxiv*. 2026
- P3. Brian M. Cho, **Raaz Dwivedi**, Nathan Kallus, “GAAVI: Global asymptotic anytime-valid inference for the conditional mean function”, *arxiv*. 2026
- P4. Kyuseong Choi, Dwaipayan Saha, Woojeong Kim, Anish Agarwal, **Raaz Dwivedi**, “GOPO: Policy optimization using ranked rewards”, *arxiv*. 2026

- P5. Tathagata Sadhukhan, Manit Paul, **Raaz Dwivedi**, “Adaptively-weighted nearest neighbors for matrix completion”, *arxiv*. 2025
- P6. Jacob Feitelberg, Dwaipayan Saha, Kyuseong Choi, Zaid Ahmad, Anish Agarwal, **Raaz Dwivedi**, “One pipeline, many transformers: Pattern-specific imputation specialists for tabular missing data”, *arxiv*. 2025
- P7. Navonil Deb, **Raaz Dwivedi**, Sumanta Basu, “Counterfactual forecasting for panel data”, *arxiv*. 2025
- P8. Jacob Feitelberg, Kyuseong Choi, Anish Agarwal, **Raaz Dwivedi**, “Distributional matrix completion via nearest neighbors in the Wasserstein space”, *arxiv*. 2024
- P9. Kyuseong Choi, Jacob Feitelberg, Anish Agarwal, **Raaz Dwivedi**, “Learning counterfactual distributions via kernel nearest neighbors”, *arxiv*. 2024
- P10. Alberto Abadie[†], Anish Agarwal, **Raaz Dwivedi**, Abhin Shah, “Doubly Robust Inference in Causal Latent Factor Models”, *arxiv*. 2024
- P11. **Raaz Dwivedi**, Katherine Tian, Sabina Tomkins, Predrag Klasnja, Susan Murphy, Devavrat Shah, “Doubly robust nearest neighbors in factor models”, *arxiv*. 2023
- P12. Abhin Shah, **Raaz Dwivedi**, Devavrat Shah, Greg Wornell, “On counterfactual inference with unobserved confounding”, *arxiv*. 2022

CONFERENCE
PUBLICATIONS

- C1. Annabelle Michael Carrell, Albert Gong, Abhishek Shetty, **Raaz Dwivedi**, Lester Mackey, “Low rank thinning”, *International Conference on Machine Learning (ICML)*. 2025
- C2. Tathagata Sadhukhan, Manit Paul, **Raaz Dwivedi**, “On adaptivity and minimax optimality of two-sided nearest neighbors”, *International Conference on Artificial Intelligence and Statistics (AISTATS)*. 2025
- C3. Albert Gong, Kyuseong Choi, **Raaz Dwivedi**, “Supervised Kernel Thinning”, *Advances in Neural Information Processing Systems (NeurIPS)*. 2024
- C4. Lingxiao Li, **Raaz Dwivedi**, Lester Mackey, “Debiased Distribution Compression”, *International Conference on Machine Learning (ICML)*. 2024
- C5. Carles Domingo-Enrich, **Raaz Dwivedi**, Lester Mackey, “Compress then test: Powerful kernel testing in near-linear time”, *Conference on Artificial Intelligence and Statistics (AISTATS)*. 2023
- C6. **Raaz Dwivedi**, Lester Mackey. “Generalized kernel thinning”, *International Conference on Learning Representations (ICLR)*. 2022
- C7. Abhishek Shetty, **Raaz Dwivedi**, Lester Mackey. “Distribution compression in near-linear time”, *International Conference on Learning Representations (ICLR)*, **Best Student Paper Award, JSM**. 2022
- C8. **Raaz Dwivedi**, Lester Mackey, “Kernel thinning”, Extended abstract in *Conference on Learning Theory (COLT)*. Full version under review in *JMLR*. 2021
- C9. **Raaz Dwivedi**^{*}, Nhat Ho^{*}, Koulik Khamaru^{*}, Martin J. Wainwright, Michael I. Jordan, Bin Yu, “Sharp analysis of Expectation-Maximization for weakly identifiable models”, *The 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*. 2020
- C10. **Raaz Dwivedi**^{*}, Nhat Ho^{*}, Koulik Khamaru^{*}, Martin J. Wainwright, Michael I. Jordan, “Theoretical guarantees for EM under misspecified Gaussian mixture models”, *Advances in Neural Information Processing Systems (NeurIPS)*. 2018
- C11. **Raaz Dwivedi**^{*}, Yuansi Chen^{*}, Martin J. Wainwright, Bin Yu, “Log-concave sampling: Metropolis-Hastings algorithms are fast”, Extended abstract in *Conference on Learning Theory (COLT)*. 2018
- C12. Yuansi Chen^{*}, **Raaz Dwivedi**^{*}, Martin J. Wainwright, Bin Yu, “Vaidya walk: A sampling algorithm based on the volumetric barrier”, *Allerton Conference*. 2017
- C13. **Raaz Dwivedi**, Vivek Borkar, “Removing sampling bias in networked stochastic approximation”, *International Conference on Signal Processing and Communications (SPCOM)*. 2014

- J1. **Raaz Dwivedi**, Katherine Tian, Sabina Tomkins, Predrag Klasnja, Susan Murphy, Devavrat Shah, “Counterfactual inference in sequential experimental design”, *Annals of Statistics (AoS)*. 2025
- J2. Carles Domingo-Enrich, **Raaz Dwivedi**, Lester Mackey, “Cheap permutation testing”, *In major revision at Annals of Statistics (AoS)*. 2025
- J3. Nhat Ho, Koulik Khamaru, **Raaz Dwivedi**, Martin J. Wainwright, Michael I. Jordan, Bin Yu, “Instability, computational efficiency, and statistical accuracy”, *Journal of Machine Learning Research (JMLR)*. 2025
- J4. **Raaz Dwivedi**, Lester Mackey, “Kernel thinning”, Extended abstract in *Journal of Machine Learning Research (JMLR)*. 2024
- J5. Raphael Kim, Susobhan Ghosh, Prasidh Chhabria, **Raaz Dwivedi**, Peng Liao, Kelly Zhang, Predrag Klasnja, Susan Murphy, “Did we personalize? Assessing personalization by an online reinforcement learning algorithm using resampling”, *Machine Learning Journal*. 2024
- J6. **Raaz Dwivedi**^{*}, Chandan Singh^{*}, Bin Yu, Martin J. Wainwright, “Revisiting minimum description length complexity in overparameterized models”, *JMLR*. 2023
- J7. Nick Altieri[†], Rebecca L. Barter, James Duncan, **Raaz Dwivedi**, Karl Kumbier, Xiao Li, Robert Netzorg, Briton Park, Chandan Singh, Yan Shuo Tan, Tiffany Tang, Yu Wang, Chao Zhang, Bin Yu, “Curating a COVID-19 data repository and forecasting county-level death counts in the United States”, *Harvard Data Science Review (HDSR)*. 2021
- J8. **Raaz Dwivedi**^{*}, Yan Shuo Tan^{*}, Briton Park, Mian Wei, Kevin Horgan, David Madigan, Bin Yu, “Stable discovery of interpretable subgroups via calibration in causal studies”, *Int. Statistical Review*. 2020
- J9. **Raaz Dwivedi**^{*}, Nhat Ho^{*}, Koulik Khamaru^{*}, Martin J. Wainwright, Michael I. Jordan, Bin Yu, “Singularity, misspecification, and the convergence rate of EM”, *Annals of Statistics (AoS)*. 2020
- J10. Yuansi Chen, **Raaz Dwivedi**, Martin J. Wainwright, Bin Yu, “Fast mixing of Metropolized Hamiltonian Monte Carlo: Benefits of multi-step gradients”, *Journal of Machine Learning Research (JMLR)*. 2020
- J11. **Raaz Dwivedi**^{*}, Yuansi Chen^{*}, Martin J. Wainwright, Bin Yu, “Log-concave sampling: Metropolis-Hastings algorithms are fast”, *Journal of Machine Learning Research (JMLR)*. 2019
- J12. **Raaz Dwivedi**[†], Ohad N. Feldheim, Ori Gurel-Gurevich, Aaditya Ramdas. “The power of online thinning in reducing discrepancy”, *Probability Theory and Related Fields (PTRF)*. 2019
- J13. Yuansi Chen^{*}, **Raaz Dwivedi**^{*}, Martin J. Wainwright, Bin Yu. “Fast MCMC sampling algorithms on polytopes”, *Journal of Machine Learning Research (JMLR)*. 2018
- J14. Vivek Borkar[†], **Raaz Dwivedi**, Neeraja Sahasrabudhe. “Gaussian approximations in high dimensional estimation”, *Systems & Control Letters*. 2016

- S1. Caleb Chin, Aashish Khubchandani, Harshvardhan Maskara, Kyuseong Choi, Jacob Feitelberg, Albert Gong, Manit Paul, Tathagata Sadhukhan, Anish Agarwal, **Raaz Dwivedi**, “N²: A unified python package and test bench for nearest neighbor-based matrix completion”, *CODEML Workshop, ICML*. 2025
- S2. Carles Domingo-Enrich, **Raaz Dwivedi**, Lester Mackey. Python package “Compress then test” ([🔗 link](#)).
- S3. Abhishek Shetty^{*}, **Raaz Dwivedi**^{*}, Lester Mackey. Python package “Compress++” ([🔗 link](#)).
- S4. **Raaz Dwivedi**, Lester Mackey. Python package “Kernel Thinning” ([🔗 link](#)).
- S5. **Raaz Dwivedi**^{*}, Yan Shuo Tan^{*}, Briton Park, Mian Wei, Kevin Horgan, David Madigan, Bin Yu. Python repository “StaDISC” ([🔗 link](#)).
- S6. Yuansi Chen^{*}, **Raaz Dwivedi**^{*}, Martin Wainwright, Bin Yu. Python package (with C++ implementation) “Vaidya and John walks” ([🔗 link](#)).

SELECTED INVITED
TALKS

AI Agents for Root Cause Analysis

- MIT Professional Education class *Feb 2026*
- Bridgewater AI Group *Oct 2025*
- Causal inference workshop, ACM Sigmetrics *Aug 2025*

Alumni Panel, 70th Anniversary Statistics Department, BSTARS 2025, UC Berkeley

Sep 2025

Integrating Double Robustness Into Causal Latent Factor Models

- Informs, Seattle *Oct 2024*
- Joint Statistical Meeting (JSM), Toronto *Aug 2024*
- ESIF conference on Economics and AI+ML *July 2024*
- Mini Workshop on Individualized Decisions, Simons/UC Berkeley *July 2024*
- Tom Ten Have Symposium on Mental Health Statistics, Weil Cornell *Jun 2024*
- Online Causal Inference Seminar *May 2024*
- Operations Research Seminar, MIT *Apr 2024*
- Workshop on Statistical Methods for Digital Health, John Hopkins University *Mar 2024*
- Statistics Seminar, Columbia *Mar 2024*
- Rising Stars in AI, KAUST *Feb 2024*

From HeartSteps to HeartBeats: Personalized Decision-making

- Large Scale Learning and Control Workshop, IIT Bombay *Dec 2023*
- AI Seminar, Cornell University *Sep 2023*
- ORIE Industry and Data Science Summit, Cornell University *Sep 2023*
- Statistics and Data Science Seminar, Cornell University *Sep 2023*
- Center for Applied Math Colloquium, Cornell University *Sep 2023*
- Gatsby Unit Seminar, University College London *Feb 2023*
- Statistics and Data Science Seminar, Yale University *Feb 2023*
- Computer Science Seminar, UIUC *Feb 2023*
- Statistics Seminar, UW Madison *Jan 2023*
- Operations, Information, and Technology Seminar, GSB, Stanford University *Jan 2023*
- Statistics and Data Science Seminar, Wharton, University of Pennsylvania *Jan 2023*
- Statistics Seminar, University of Chicago *Jan 2023*
- Statistics and Operation Research Seminar, UNC Chapel Hill *Jan 2023*
- Statistics Seminar, UCLA *Jan 2023*
- Operation Research and Industrial Engineering Seminar, Cornell University *Dec 2022*
- Operation Research and Industrial Engineering Seminar, Cornell Tech *Dec 2022*
- Statistics Seminar, Rutgers University *Nov 2022*
- ISL Colloquium, EE, Stanford University *Nov 2022*
- BLISS Seminar, EECS, UC Berkeley *Nov 2022*

Compress then test: Powerful kernel testing in near-linear time

- Joint Statistical Meeting, Toronto *Jun 2023*
- Monte Carlo Methods Conference, Paris *Jun 2023*
- Computational-Statistical Interplay in Machine Learning Workshop, MIT *May 2023*

Doubly robust nearest neighbors for counterfactual inference

- Causal Inference Workshop, ACM Sigmetrics, Orlando *Jun 2023*

	• New England Statistics Symposium, Boston University • Informs Annual Meeting, Indianapolis	<i>Jun 2023</i> <i>Oct 2022</i>
	Counterfactual inference in sequential experiments	
	• Institute of Mathematical Statistics (IMS) Annual Meeting, London • Learning from Interventions Workshop, Simons Institute, Berkeley	<i>Jun 2022</i> <i>Feb 2022</i>
	Near-optimal compression in near-linear time	
	• SIAM Conference on Uncertainty Quantification, Atlanta • Statistical learning Workshop, Mathematical Sciences Research Institute, Berkeley	<i>Apr 2022</i> <i>Mar 2022</i>
	Kernel thinning	
	• Data-Centric Engineering Group, Alan Turing Institute, Virtual	<i>Sep 2021</i>
	Revisiting minimum description length complexity in overparameterized models	
	• Alg. Info Theory & Machine Learning Symp., Alan Turing Institute, London • Collaborations on the Theoretical Foundations of Deep Learning, Virtual	<i>Jul 2022</i> <i>Nov 2021</i>
	StaDISC: Stable discovery of interpretable subgroups via calibration	
	• Young Data Scientist Research Seminar, ETH Zurich, Virtual • ASA Annual Symposium on Data Science & Statistics, Virtual	<i>Sep 2020</i> <i>Jun 2020</i>
	Singularity, misspecification, & the convergence rate of EM	
	• Math & Statistics Seminar, IIT Kanpur • AMS Special Sections Meeting, UC Riverside	<i>Jan 2020</i> <i>Nov 2019</i>
	Theoretical guarantees for MCMC algorithms	
	• BIDS Seminar, UC Berkeley • EE Seminar, IIT Bombay • STCS Seminar, TIFR Bombay	<i>Mar 2019</i> <i>Jan 2018</i> <i>Jan 2018</i>
PHD STUDENTS	Kyuseong Choi , Fifth Year, Statistics	<i>2024–</i>
	Brian Cho , Fourth Year, ORIE	<i>2024–</i>
	Albert Gong , Third Year, CS	<i>2024–</i>
MINOR COMMITTEE MEMBER	Navonil Deb , Fifth Year, Statistics	<i>2024–</i>
	Tathagata Sadhukhan , Second Year, Statistics	<i>2024–</i>
TEACHING EXPERIENCE	T1. Statistical Principles for Data Science Inference (ORIE 5180), <i>Cornell University</i>	<i>Fall 2024</i>
	T2. Causal Inference (ORIE 7790), <i>Cornell University</i>	<i>Spring 2024</i>
	T3. Instructor: Statistical Principles (ORIE 6700), <i>Cornell University</i>	<i>Fall 2023</i>
	T4. Instructor: Statistical RL for real life (one week; link), <i>CDT Summer School, Missenden</i>	<i>Jul 2023</i>
	T5. TA: Sequential Decision Making (STAT 234), <i>Harvard University</i> . Gave four guest lectures and supervised several half-semester long research projects.	<i>Spring 2022</i>
	T6. TA: Modern Statistical Prediction and Machine Learning (STAT 154), <i>UC Berkeley</i> . Gave one guest lecture and helped in redesign of the class.	<i>Spring 2019</i>

- T7. TA: Introduction to Machine Learning (EECS 189), *UC Berkeley*. Co-head for the content developments in team of 10+ TAs, helped design discussion sections, homeworks, and exams. *Spring 2018*
- T8. TA: Linear Algebra, Calculus, Differential equations (MA 105, 106, 108, 207), *IIT Bombay*. Taught teaching sections and several voluntary help sessions that were often attended by 200+ students. *2011–2014*

Undergraduate Research Mentoring

- UC Berkeley, One student that led to a co-authored journal publication *2020–2021*
- Harvard, Two students with three co-authored submissions in preparation *2022–*

Institutional Mentoring

- MIT Institute for Data, Systems, & Society (IDSS) Postdoc Mentors for two *PhD* students *2022–*
- UC Berkeley Artificial Intelligence Research (BAIR) Buddies for two *incoming PhD* students *2020–2021*
- UC Berkeley BAIR Mentoring Program for five *undergraduates* *2017–2021*
- IIT Bombay Student Mentoring Program (ISMP) for twelve *incoming undergraduates* *2013–2014*
- IIT Bombay Academic Mentoring Program (DAMP) for four *sophomores & juniors* *2012–2014*
- IIT Bombay Intensive Mentoring Program for thirty *undergraduates* *2012–2013*

Committees

- Member, Committee on Equality and Diversity, IMS *2022–*

Scientific Meetings

- Organizer and chair, Informs Session on Reinforcement learning *2024*
- Organizer and chair, Informs Session on Causal inference *2024*
- Organizer and chair, Informs Session on Causal inference and reinforcement learning *2023*
- Organizer and chair, Informs Session on Statistical Methods for Healthcare *2023*
- Mentor, Let-All Mentoring Session, Learning Theory Mentorship Workshop *2023*
- Moderator, Panel Discussion on Mentoring, New Researcher Conference Statistics, Toronto *2023*
- Chair, New Researchers Group Session, IMS Annual Meeting *2022*
- Chair, Statistical Machine Learning Session, IMS Annual Meeting *2022*
- Mentor, Summer Institute on Just-in-Time Adaptive Interventions via MRTs *2021*

Graduate Admissions

- ORIE Graduate Admissions Committee, Cornell *2024*
- ORIE Graduate Admissions Committee, Cornell *2023*
- EECS Graduate Admissions Committee, MIT *2021*
- EECS Graduate Admissions Committee, UC Berkeley *2018–2020*

Reviewing Activities

- *Journals*: MOR, OR, Sto. Sys., AOS, JMLR, IEEE-IT, JRSSB, Bernoulli, HDSR, Stats-Comp., SIAM, Jour. of Causal Inference, ISR, JCGS, ACM
- *Conferences*: NeurIPS (Area Chair), COLT, ICML, AISTATS, FOCS, STOC, SODA, AAAI, UAI, SIGMETRICS

WORK EXPERIENCE	Microsoft Research , Research Intern with Lester Mackey, New England, USA	2019
	Mist Systems, Juniper Networks, Data Science Intern, Cupertino, USA	2017
	WorldQuant Research, Senior Quantitative Researcher, Mumbai, India	2014–2015
	Stanford University , Research Intern with Prof. Balaji Prabhakar, USA	2013
	Ivy Mobility, Data Science Intern, Chennai, India	2012